Assignment No-04

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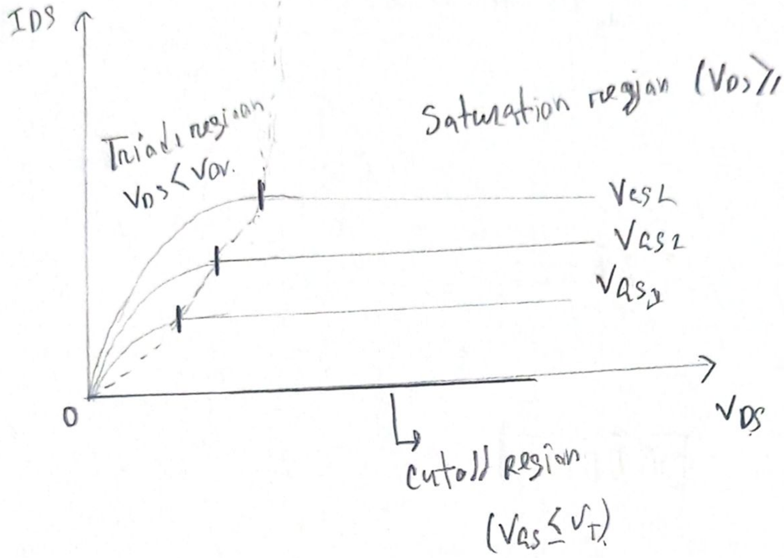
Course : CSE251

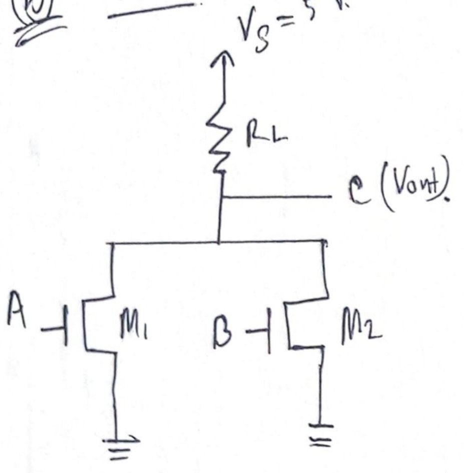
Section : 22.

Ans to the Q No-1

(a) I-V characteristics of Mosfet.

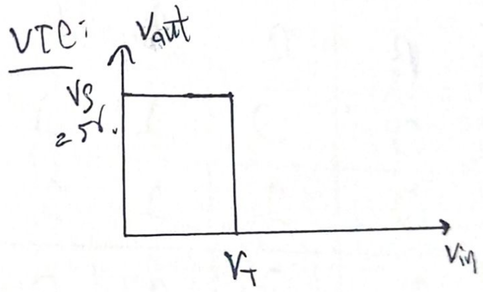


(b) NOR:



when A = 0
C = 1 for B = 0
C = 0 for B = 1

VTC:



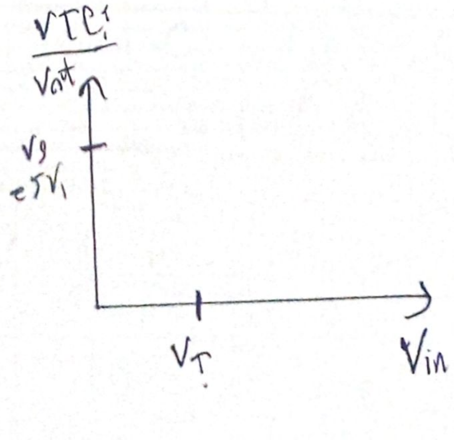
Truth table:

A	B	C (Vout)
0	0	1 (5V)
0	1	0
1	0	0
1	1	0

when, A = 1

C = 0 for B = 0 and B = 1

VTC:



Ans to the Q No-2

(a) $Z = \overline{(A + \bar{A}B + BC)}$

$= \bar{A}\,(\bar{\bar{A}}+\bar{B})(\bar{B}+\bar{C})$ [De' morgan's law]

$= \bar{A}\,(A+\bar{B})(\bar{B}+\bar{C})$

$= (A\bar{A}+\bar{A}\bar{B})(\bar{B}+\bar{C})$

$= (0+\bar{A}\bar{B})(\bar{B}+\bar{C})$ [$A\bar{A}=0$]

$= \bar{A}\bar{B}\bar{B} + \bar{A}\bar{B}\bar{C}$

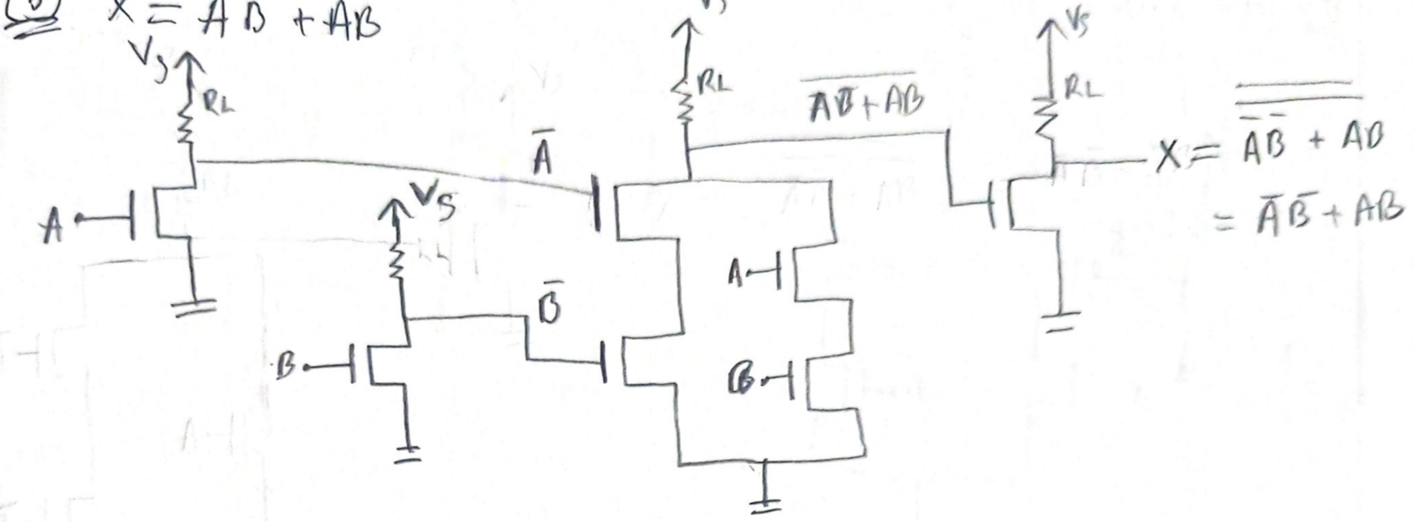
$= \bar{A}\bar{B} + \bar{A}\bar{B}\bar{C}$ [$\bar{B}\bar{B}=\bar{B}$]

$\therefore Z = \bar{A}\bar{B}$ [$A+AB=A$]

Truth table:

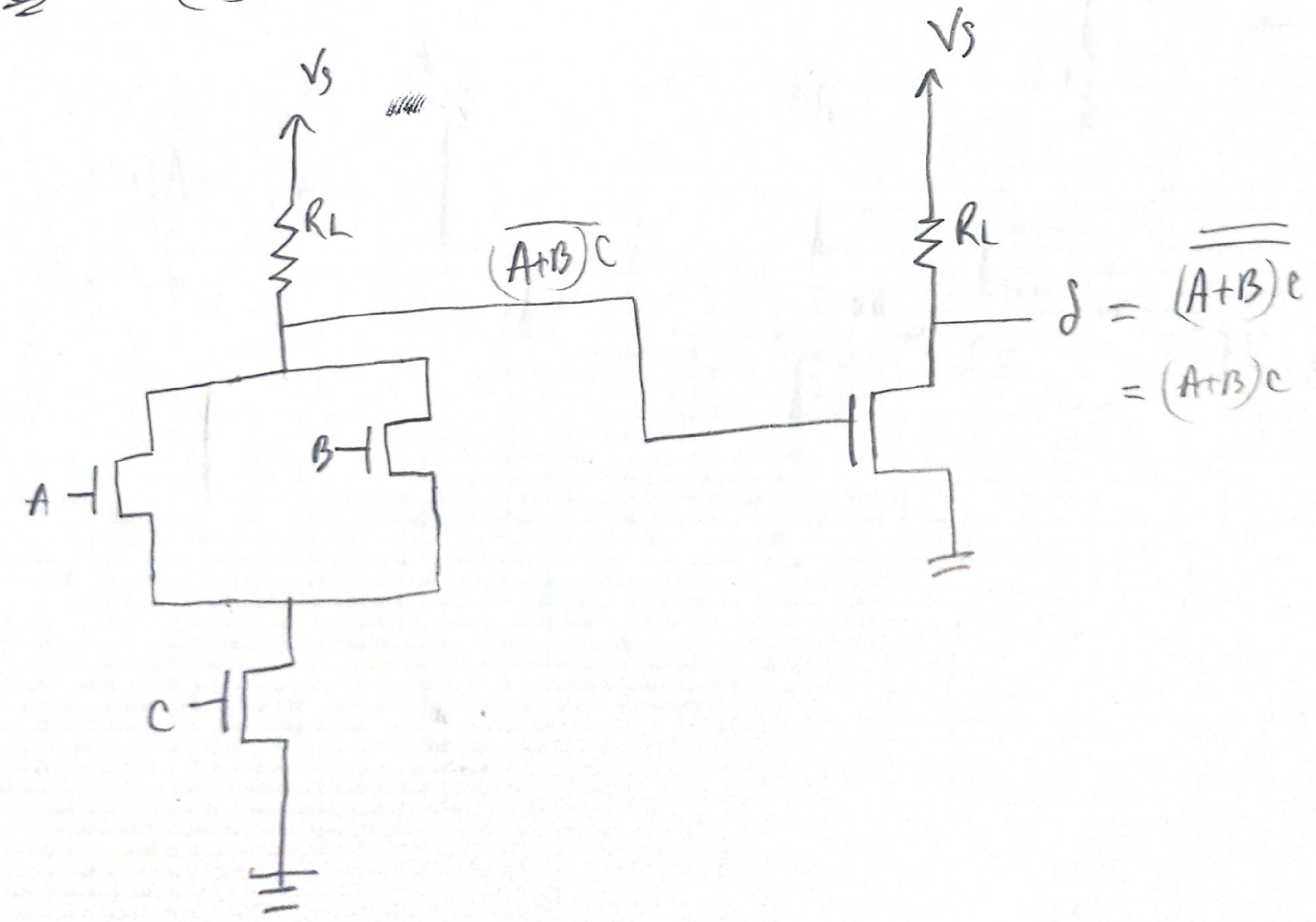
A	B	C	$\bar{A}$	$\bar{B}$	Output
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	0	1	0
1	1	0	0	0	0
1	1	1	0	0	0

(b) $X = \bar{A}\bar{B} + AB$

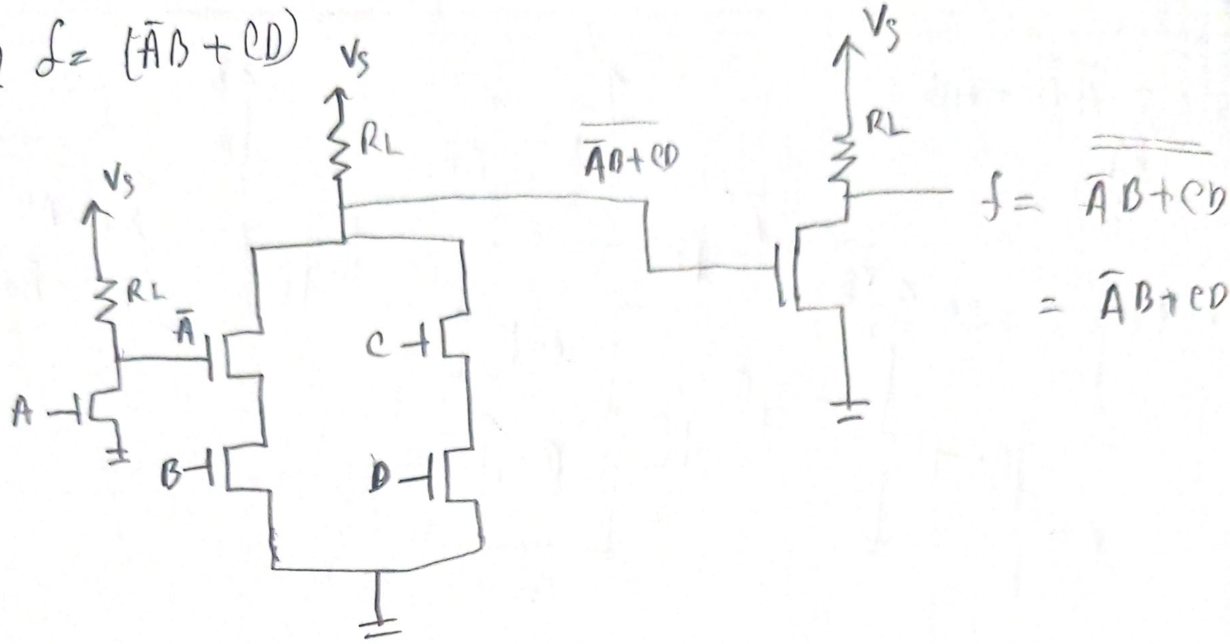


Ans to the Q No-3

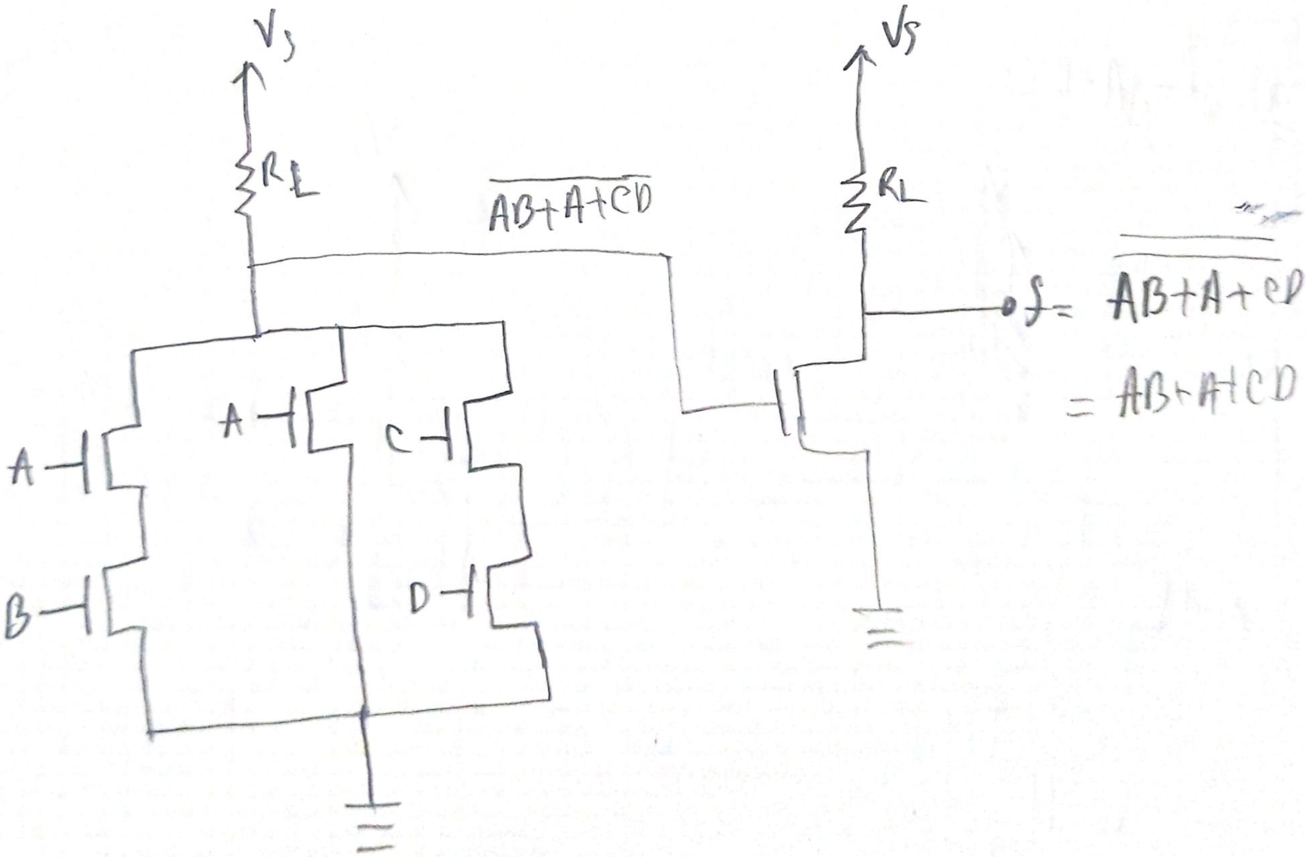
(a) $\delta = \overline{(A+B)C}$



(b) $f = (\bar{A}B + CD)$



(c) $f = AB + A + CD.$



Here, Ans to the Q No-4

$V_t = 1$, $V_G = 4V$, $i_{DS} = 2mA$ $k = 4mA/V^2$

so $i_{DS} = \frac{10 - V_D}{1}$

$\Rightarrow V_D = 10 - 2 = 8V$,

Assume the MOSFET is in saturation mode, so, $V_{DS} \geq V_{OV}$

$i_{DS} = \frac{k}{2} [V_{GS} - V_t]^2$,

$\Rightarrow 2 = \frac{4}{2} [V_G - V_S - V_t]^2$

$\Rightarrow 2 = 2 [4 - V_S - 1]^2$

$\Rightarrow V_S^2 - 6V_S - 8 = 0$

$\Rightarrow V_S = 4$ or 2.

$V_S = 4$ [not possible] so, $V_S = 2V$.

Verify :-

$V_{DS} = V_D - V_S = 8 - 2 = 6V$

$V_{GS} = V_G - V_S = 2V$,

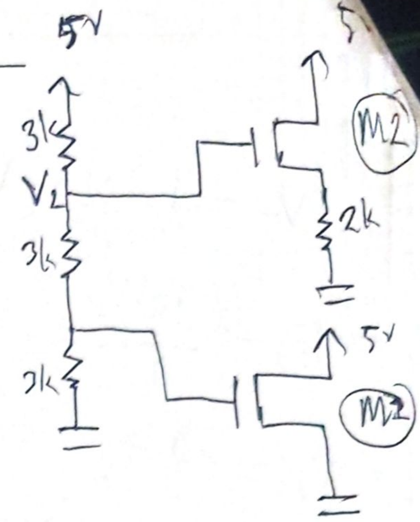
$V_{OV} = 2 - 1 = 1V$.

Here $V_{DS} \geq V_{OV}$ so the mosfet is in saturation mode.

∴ Assumption correct.

Ans

Ans to the Q No-5



(i) $E=2$, $V_T=0.3$, $K_n=0.1\ mA/V^2$

For M1:

$$V_1=\frac{3}{9}\times 5 \quad \text{[Voltage divider rule]}$$

$$=1.67V$$

$V_G=1.67V$, $V_S=0V$

$V_{GS}=1.67V$

$\therefore \boxed{V_{GS}>V_T}$

Here, $V_D=5V$.

$V_{DS}=5V$

Now,

$V_{OV}=V_{GS}-V_T=1.67-0.3=1.37V$

Here $V_{DS}\geq V_{OV}$

$\therefore$ M1 is in saturation mode,

$$\therefore i_{DS}=\frac{k}{2}[V_{OV}]^2=\frac{0.1}{2}[1.37]^2=0.0938\ mA.$$

Ans

For M2:

$V_2=\frac{6}{9}\times 5=3.33\ V$ [Voltage divider rule]

$\therefore V_G=3.33V$, $V_D=5V$, $V_T=0.3V$ $k=0.1\ mA/V^2$

Let's assume M2 is in saturation mode,

We know

$i_{DS} = \frac{k}{2}[V_{GS} - V_T]^2$

$i_{DS} = i_S = i_{OS}$

$i_S = \frac{V_S - 0}{2} = \frac{V_S}{2}$

$\frac{V_S}{2} = \frac{0.1}{2}[V_G - V_S - V_T]^2$

$\Rightarrow \frac{V_S}{2} = [3.33 - V_S - 0.3]^2 \times \frac{0.1}{2}$

$\Rightarrow V_S = (9.18 - 6.06V_S + V_S)^2 \times 0.1$

$V_S = 0.918 - 0.606V_S + 0.1V_S^2$

$\Rightarrow 0.1V_S^2 - 1.606V_S + 0.918 = 0$

$\Rightarrow V_S = 15.46\ V$ or $0.59\ V$

So $V_S = 0.59\ V$. [$V_S = 15.46$ not possible]

$\therefore V_{GS} = 3.33 - 0.5 = 2.74\ V$

$\therefore \boxed{V_{GS} > V_T}$

Now $V_{DS} = V_D - V_S = 5 - 0.59 = 4.41\ V$

$V_{OV} = V_{GS} - V_T = 2.74 - 0.3 = 2.44\ V$

$\therefore \boxed{V_{DS} > V_{OV}}$

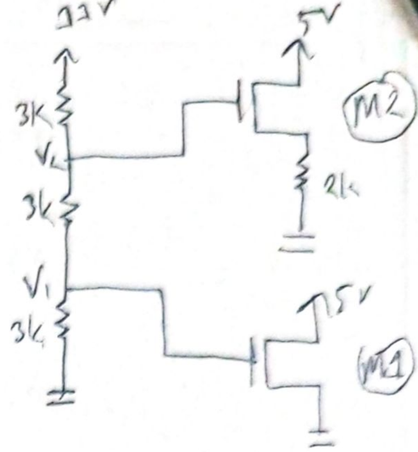
So assumption correct. It is in saturation mode.

$\therefore P_{2k\Omega} = V_S I_S$

$= 0.59 \times \frac{0.59}{2}$

$= 0.175\ mW$

Ans



iii) $E=5$, $V_T=0.3V$, $K_n=0.1mA/V^2$.

For M1:

$$V_1=\frac{3}{9}\times 11=3.67V \quad \text{[voltage divider rule]}$$

$\therefore V_G=3.67V$, $V_S=0V$

$\therefore V_{GS}=3.67V$

$\therefore \boxed{V_{GS}>V_T}$

Now, $V_D=5V$.

$V_{DS}=5V$, $\quad V_{ov}=V_{GS}-V_T=3.67-0.3=3.37V$.

So, $\boxed{V_{DS}>V_{ov}}$

$\therefore$ M1 mosfet is in saturation mode.

$$\therefore I_{DS}=\frac{K}{2}[V_{ov}]^2=\frac{0.1}{2}[3.37]^2=0.57mA.$$

For M2

$$V_2=\frac{6}{9}\times 11=7.33V.$$

$\therefore V_G=7.33V$, $V_D=5V$, $V_T=0.3V$, $K_n=0.1\ mA/V^2$. Let's assume

M2 mosfet in in Triode mode.

$$\Big|\ i_{DS}=\frac{V_S}{2}.$$

So $i_{DS}=K\left[V_{GS}-V_T-\frac{1}{2}V_{DS}\right]V_{DS}$

$$\Rightarrow i_{DS}=K\left(V_G-\frac{1}{2}V_S-V_T-\frac{1}{2}V_D\right)\cdot(V_D-V_S)$$

$$\Rightarrow i_{DS}=K\left(7.33-\frac{1}{2}V_S-0.3-\frac{1}{2}5\right)(5-V_S)$$

$$\Rightarrow i_{DS}=K\left(22.65-\frac{5}{2}V_S-4.53V_S+\frac{1}{2}V_S^2\right)$$

$$\Rightarrow \frac{V_S}{2}=0.1\left(\frac{1}{2}V_S^2-7.03V_S+22.65\right)$$

$\therefore V_S = 0.2\left(\frac{1}{2}V_S^2 - 7.03V_S + 22.65\right)$

$\therefore 0.1V_S^2 - 2.406V_S + 4.53 = 0$

So, $V_S = 22V$ or $V_S = 2.05V$. [$V_S = 22V$ not possible]

So $V_{GS} = 7.33 - 2.05 = 5.28\,V$ [$\because V_{GS} > V_T$]

$V_{DS} = 5 - 2.05 = 2.95\,V$

$V_{OV} = 5.28 - 0.3 = 4.98\,V$

$\therefore V_{DS} < V_{OV}$ [Assumption is correct]

$P_{2k\Omega} = V_S I_S = \left(2.05 \times \frac{2.05}{2}\right) = 2.11\,mW$. Ans

Ans to the Q. No. 6

(a) Condition to verify the saturation mode of a MOSFET are

i) $V_{GS} > V_T$ and ii) $V_{DS} \geq V_{OV}$.

(b) Here, $V_G = 5\,V$ $V_T = 1\,V$, $V_S = 0\,V$, $k = 4\,mA/V^2$

$V_{GS} = 5V$. So, $\boxed{V_{GS} > V_T}$

Let's assume Mosfet is in saturation mode;

$i_{DS} = \frac{10 - V_D}{0.1}$

$I_{DS} = \frac{k}{2}[V_{OV}]^2$

$= \frac{4}{2}[5 - 0 - 1]^2$

$\therefore \frac{10 - V_D}{0.1} = \frac{4}{2}$

$\therefore V_D = 10 - 3.2 = 6.8\,V$.

$\therefore V_{DS} = 6.8V.$

$V_{ov} = 5-1 = 4V.$

So, $\boxed{V_{DS} > V_{ov}}$ Assumption correct.

$\therefore V_{out} = 6.8V.$

Nirmal told the inverter will malfunction. As $V_D = 6.8 > 5$. So the output was high when input was high. So it malfunctioned. Nirmal was right.

(Ans)

(c) Here, $V_G = 5V$, $V_S = 0V$, $V_T = 1V$, $R_D = 1k\Omega$

$\therefore V_{GS} = 5V$, $V_{ov} = 5-1 = 4V.$

$\therefore \boxed{V_{GS} > V_T}$ Let's assume it's in triode mode,

$V_{DS} = V_D$

$i_{DS} = 10 - V_D$

So, $i_{DS} = K_n \left[V_{GS} - V_T - \frac{1}{2}V_{DS}\right] V_{DS}$

$\Rightarrow 10 - V_D = 4\left[5-1-\frac{1}{2}V_D\right]V_D$

$\Rightarrow 10 - V_D = 16V_D - 2V_D^2$

$\Rightarrow 2V_D^2 - 17V_D + 10 = 0$

$\therefore V_D = 7.86V$ or $V_D = 0.63V.$

We take $V_D = 0.63V$,

$\therefore V_{DS} = 0.63V.$

$\therefore V_{DS} < V_{ov}$ [Assumption correct]

The inverter in working properly after modification.

(Ans)

(d) $V_G = 5$, $V_S = 0$, $V_T = 1$,

$V_{ov} = 5 - 1 = 4V$.

At the edge of saturation, $V_{DS} = V_{ov}$.

$\therefore V_{DS} = 4V$, so, $V_{DS} = V_D = 4V$.

Now

$i_{DS} = \frac{k}{2} [V_{ov}]^2 = \frac{4}{2} [4]^2 = 32 \, mA$

so,

$i_{DS} = \frac{10 - V_D}{R_D}$

$R_D = \frac{10 - V_D}{i_{DS}}$

$R_D = \frac{10 - 4}{32 \times 10^{-3}}$

$R_D = 187.5 \, \Omega$

$R_D = 0.1875 \, k\Omega$.

$\therefore R_D = 0.1875 \, k\Omega$ the mosfet operates at the

$\therefore$ When $R_D = 0.1875 \, k\Omega$ the mosfet operates at the edge of saturation.

(Ans)